Biostat 200C Homework 3

Due May 9 @ 11:59PM

2025-05-09

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0.1 Q1. Concavity of Poisson regression log-likelihood (50pts)

Let $Y_1, \ldots, Y_n$ be independent random variables with $Y_i \sim \text{Poisson}(\mu_i)$ and $\log \mu_i = \mathbf{x}_i^T \beta$, $i = 1, \ldots, n$.

0.1.1 Q1.1 (10pts)

Write down the log-likelihood function.

0.1.2 Q1.2 (10pts)

Derive the gradient vector and Hessian matrix of the log-likelihood function with respect to the regression coefficients β.

0.1.3 Q1.3 (20pts)

Show that the log-likelihood function of the log-linear model is a concave function in regression coefficients β. (Hint: show that the negative Hessian is a positive semidefinite matrix.)

0.1.4 Q1.4 (10pts)

Show that for the fitted values $\hat{\mu}_i$ from maximum likelihood estimates

$$\sum_i \hat{\mu}_i = \sum_i y_i.$$

Therefore the deviance reduces to

$$D = 2 \sum_i y_i \log \frac{y_i}{\hat{\mu}_i}.$$

0.2 Q2. Show negative binomial distribution mean and variance (20pts)

Recall the probability mass function of negative binomial distribution is

$$\mathbb{P}(Y = y) = \binom{y + r - 1}{r - 1} (1 - p)^r p^y, \quad y = 0, 1, \ldots$$

Show $\mathbb{E}Y = \mu = rp/(1 - p)$ and $\operatorname{Var} Y = rp/(1 - p)^2$.

0.3 Q3. ELMR Chapter 5 Exercise 5 (page 100) (70pts)

The dvisits data comes from the Australian Health Survey of 1977–1978 and consist of 5190 single adults where young and old have been oversampled.
(a) (5pts)
Make plots which show the relationship between the response variable, doctorco, and the potential predictors, age and illness.

```
library(faraway)
```

```
Warning: package 'faraway' was built under R version 4.3.3
```

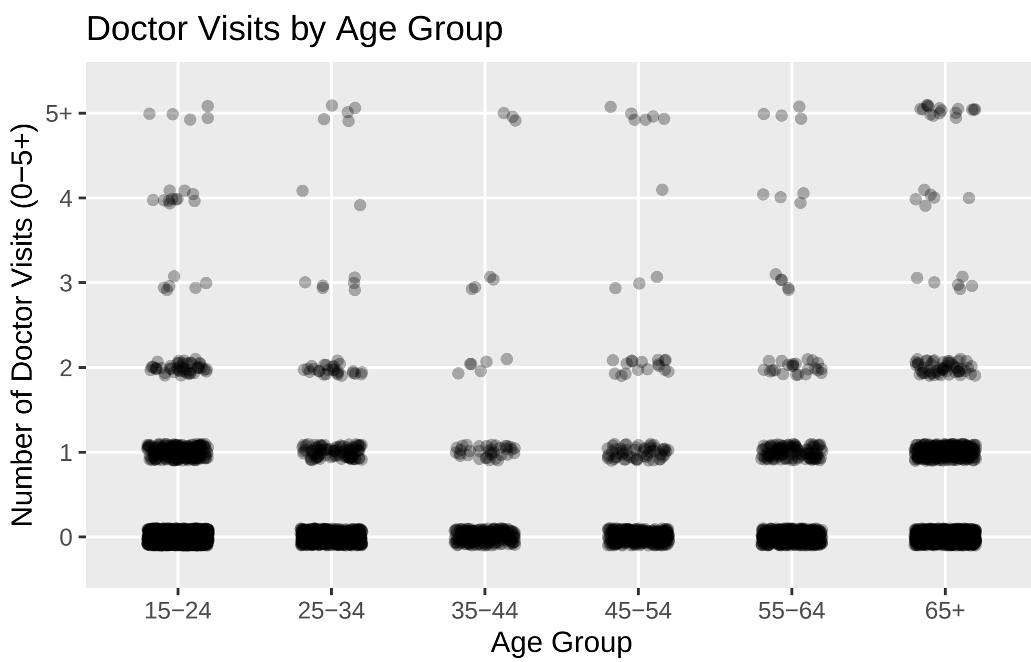
```
library(ggplot2)
```

```
Warning: package 'ggplot2' was built under R version 4.3.1
```

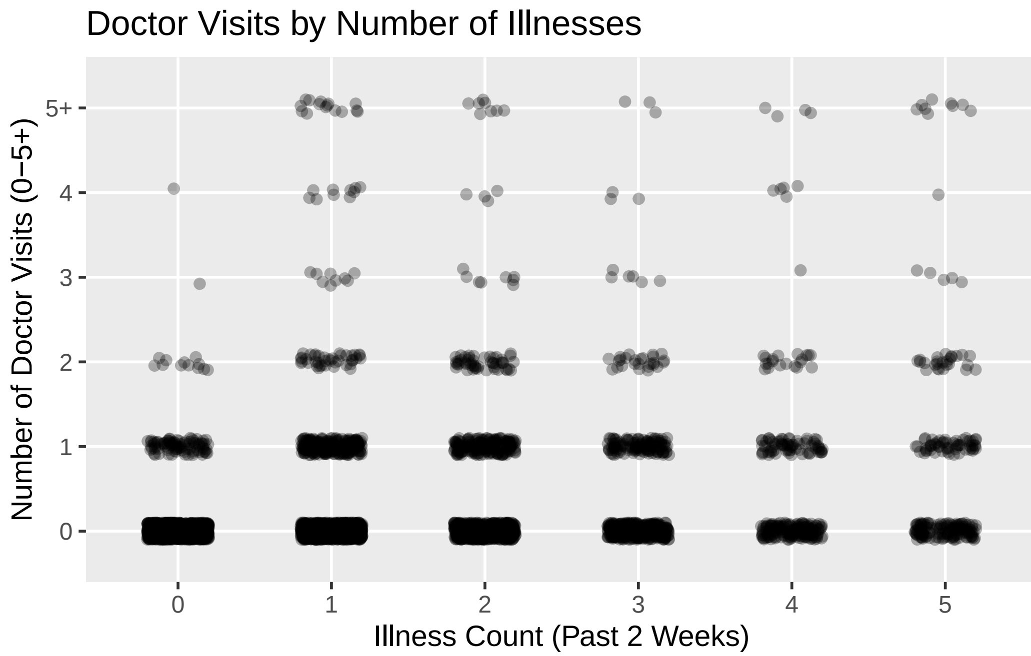
```r
data(dvisits)

library(dplyr)
```

```
Warning: package 'dplyr' was built under R version 4.3.1
```

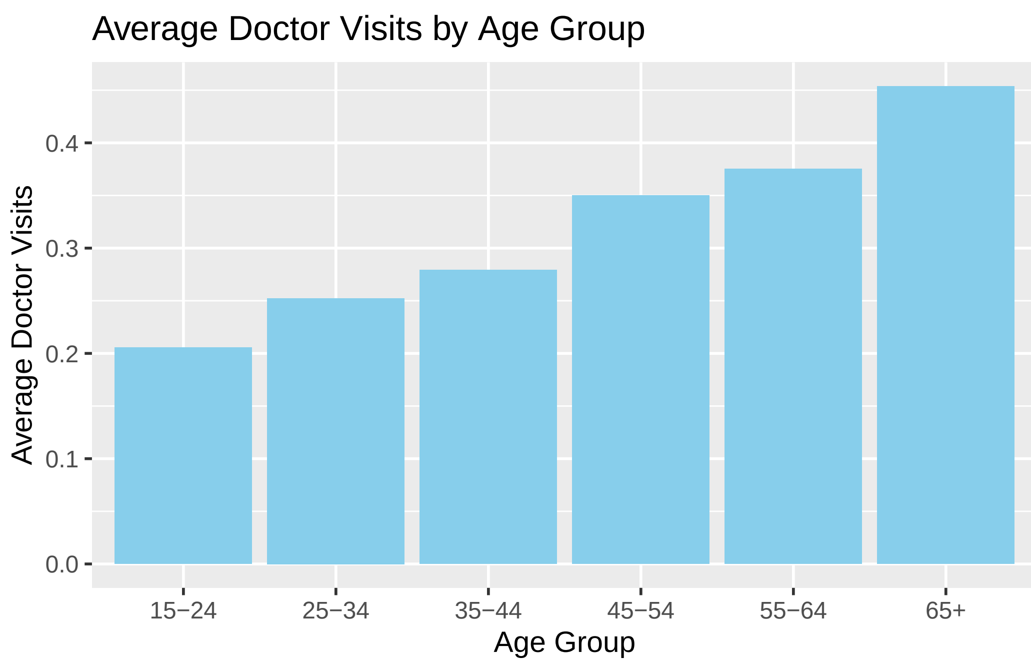
```r
# Dividing age into groups for better visualization
dvisits <- dvisits %>%
  mutate(age_group = cut(age*100, breaks = c(15, 25, 35, 45, 55, 65, 75),
                         labels = c("15-24", "25-34", "35-44", "45-54", "55-64",
                                    "65+")))

# Scatter plot: doctorco vs. age_group
dvisits <- dvisits %>%
  mutate(
    doctorco_plot = ifelse(doctorco >= 5, 5, doctorco),
    doctorco_plot = factor(doctorco_plot, levels = 0:5,
                           labels = c("0", "1", "2", "3", "4", "5+"))
  )

ggplot(dvisits, aes(x = age_group, y = doctorco_plot)) +
  geom_jitter(width = 0.2, height = 0.1, alpha = 0.3) +
  scale_y_discrete(limits = c("0", "1", "2", "3", "4", "5+")) +
  labs(title = "Doctor Visits by Age Group",
       x = "Age Group",
       y = "Number of Doctor Visits (0-5+)")
```



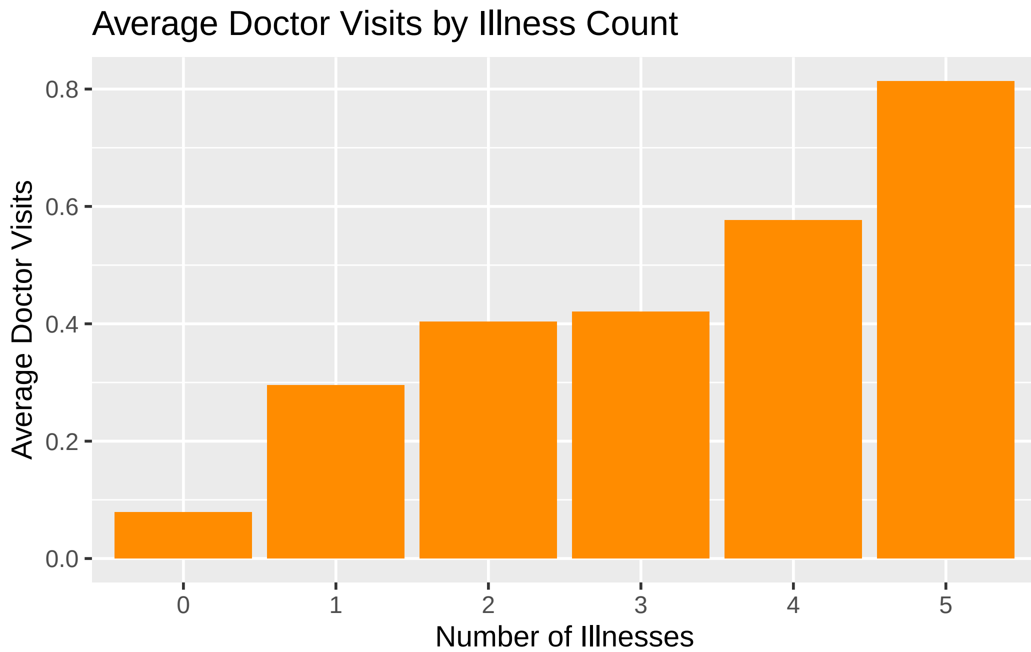

```r
# Scatter plot: doctorco vs. illness
ggplot(dvisits, aes(x = factor(illness), y = doctorco_plot)) +
  geom_jitter(width = 0.2, height = 0.1, alpha = 0.3) +
  scale_y_discrete(limits = c("0", "1", "2", "3", "4", "5+")) +
  labs(title = "Doctor Visits by Number of Illnesses",
       x = "Illness Count (Past 2 Weeks)",
       y = "Number of Doctor Visits (0-5+)")
```




```
# Bar plot of average doctor visits by age group
dvisits %>%
  group_by(age_group) %>%
  summarise(avg_doctorco = mean(doctorco)) %>%
  ggplot(aes(x = age_group, y = avg_doctorco)) +
  geom_col(fill = "skyblue") +
  labs(title = "Average Doctor Visits by Age Group", x = "Age Group",
       y = "Average Doctor Visits")
```



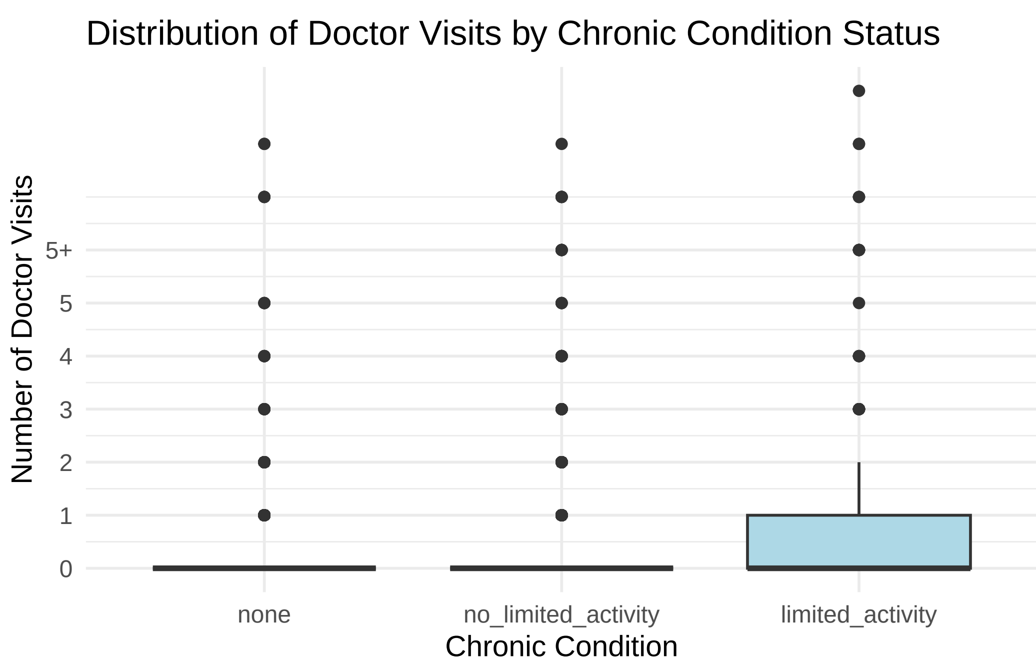

```
# Bar plot of mean doctorco by illness count
dvisits %>%
  group_by(illness) %>%
  summarise(mean_doctorco = mean(doctorco)) %>%
  ggplot(aes(x = factor(illness), y = mean_doctorco)) +
  geom_col(fill = "darkorange") +
  labs(title = "Average Doctor Visits by Illness Count",
       x = "Number of Illnesses", y = "Average Doctor Visits")
```



(b) (5pts)
Combine the predictors chcond1 and chcond2 into a single three-level factor. Make an appropriate plot showing the relationship between this factor and the response. Comment.

```
dvisits$cond <- with(dvisits,
                     ifelse(chcond1 == 0 & chcond2 == 0, "none",
                     ifelse(chcond1 == 1 & chcond2 == 0, "no_limited_activity",
                     ifelse(chcond1 == 0 & chcond2 == 1, "limited_activity",
                            NA))))

# Turn into a factor with proper ordering
dvisits$cond <- factor(dvisits$cond,
                       levels = c("none", "no_limited_activity",
                                  "limited_activity"))


dvisits$cond <- factor(dvisits$cond,
                       levels = c("none", "no_limited_activity",
                                  "limited_activity"))


library(ggplot2)
```

```r
library(dplyr)

# Cap doctorco at 5+ for clean y-axis categories
dvisits$doctorco_plot <- ifelse(dvisits$doctorco >= 5, 5, dvisits$doctorco)
dvisits$doctorco_plot <- factor(dvisits$doctorco_plot,
                                levels = 0:5,
                                labels = c("0", "1", "2", "3", "4", "5+"))


# Side-by-side boxplots
ggplot(dvisits, aes(x = cond, y = doctorco)) +
  geom_boxplot(fill = "lightblue") +
  scale_y_continuous(breaks = 0:6, labels = c("0", "1", "2", "3", "4", "5",
                                              "5+")) +
  labs(title = "Distribution of Doctor Visits by Chronic Condition Status",
       x = "Chronic Condition",
       y = "Number of Doctor Visits") +
  theme_minimal()
```



In the boxplot, individuals with no chronic conditions or with chronic conditions without activity limitation have nearly identical distributions, with almost all visits clustered at zero. In contrast, individuals with chronic conditions that limit activity show a visibly higher spread, with a greater number of visits and several high outliers. This could suggest that activity limiting chronic conditions are strongly associated with increased healthcare utilization.

(c) (10pts)
Build a Poisson regression model with doctorco as the response and sex, age, agesq, income, levyplus, freepoor, freerepa, illness, actdays, hscore and the three-level condition factor as possible predictor variables. Considering the deviance of this model, does this model fit the data?

```r
dvisits$agesq <- dvisits$age^2

full_model <- glm(doctorco ~ sex + age + agesq + income + levyplus +
                  freepoor + freerepa + illness + actdays + hscore + cond,
                  data = dvisits, family = poisson)

summary(full_model)
```

```

Call:
glm(formula = doctorco ~ sex + age + agesq + income + levyplus +
    freepoor + freerepa + illness + actdays + hscore + cond,
    family = poisson, data = dvisits)

Coefficients:
                         Estimate Std. Error z value Pr(>|z|)
(Intercept)             -2.223848   0.189816 -11.716   <2e-16 ***
sex                      0.156882   0.056137   2.795   0.0052 **
age                      1.056299   1.000780   1.055   0.2912
agesq                   -0.848704   1.077784  -0.787   0.4310
income                  -0.205321   0.088379  -2.323   0.0202 *
levyplus                 0.123185   0.071640   1.720   0.0855 .
freepoor                -0.440061   0.179811  -2.447   0.0144 *
freerepa                 0.079798   0.092060   0.867   0.3860
illness                  0.186948   0.018281  10.227   <2e-16 ***
actdays                  0.126846   0.005034  25.198   <2e-16 ***
hscore                   0.030081   0.010099   2.979   0.0029 **
condno_limited_activity  0.114085   0.066640   1.712   0.0869 .
condlimited_activity     0.141158   0.083145   1.698   0.0896 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 5634.8  on 5189  degrees of freedom
Residual deviance: 4379.5  on 5177  degrees of freedom
AIC: 6737.1

Number of Fisher Scoring iterations: 6
```

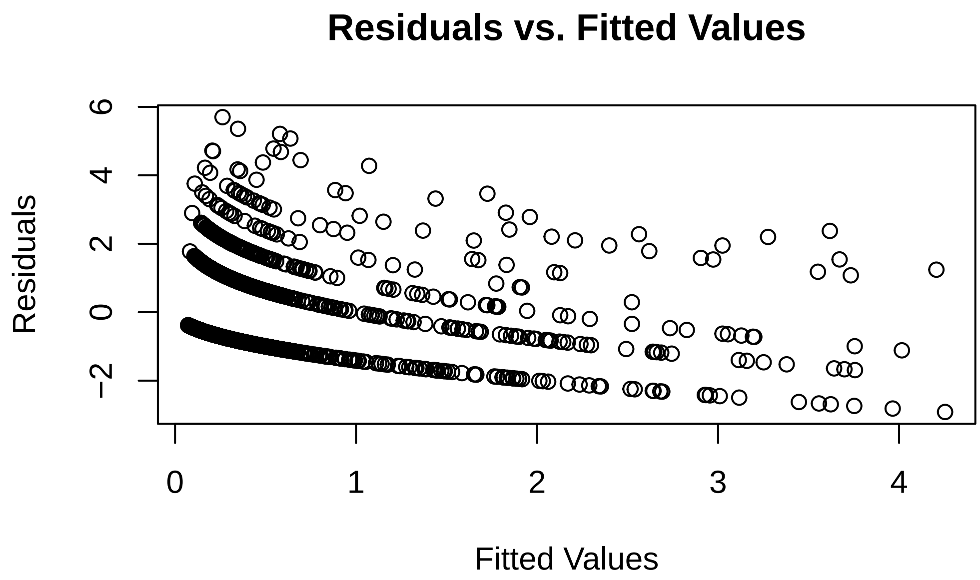
```
pchisq(full_model$deviance, full_model$df.residual, lower = FALSE)
```

```
[1] 1
```

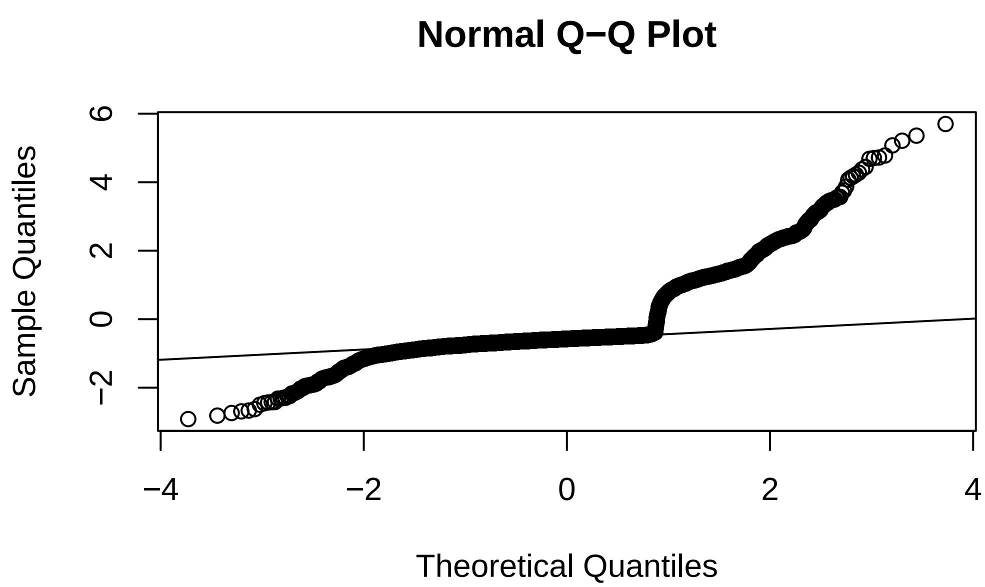
The deviance goodness-of-fit test returned a p-value of 1, indicating that the residual deviance is much smaller than expected under the Poisson model. This suggests no evidence of lack of fit. However, such a high p-value may also signal underdispersion or possible overfitting.

(d) (10pts)
Plot the residuals and the fitted values — why are there lines of observations on the plot? Make a QQ plot of the residuals and comment

```
plot(full_model$fitted.values, resid(full_model), xlab = "Fitted Values",
     ylab = "Residuals", main = "Residuals vs. Fitted Values")
```




```r
# QQ plot of residuals
qqnorm(resid(full_model))
qqline(resid(full_model))
```



The residuals vs. fitted values plot shows multiple horizontal bands which is expected for count data like doctorco, since it only takes integer values. There's a slight increase in residual variance at higher fitted values, which could suggest mild overdispersion or that variance increases with the mean which is common in Poisson data.
The QQ plot of deviance residuals shows some deviation from normality, especially in the upper tail. While this is expected with count data and Poisson regression, it suggests that the model may not fully capture the distribution of the data.

(e) (10pts)
Use a stepwise AIC-based model selection method. What sort of person would be predicted to visit the doctor the most under your selected model?

```r
step_model <- step(full_model, direction = "both")
```

```
Start:  AIC=6737.08
doctorco ~ sex + age + agesq + income + levyplus + freepoor +
    freerepa + illness + actdays + hscore + cond

           Df Deviance    AIC
- agesq     1   4380.1 6735.7
- freerepa  1   4380.3 6735.8
- age       1   4380.6 6736.2
- cond      2   4383.2 6736.7
<none>          4379.5 6737.1
- levyplus  1   4382.5 6738.1
- income    1   4385.0 6740.5
- freepoor  1   4386.2 6741.8
- sex       1   4387.4 6743.0
- hscore    1   4388.1 6743.7
- illness   1   4481.8 6837.4
- actdays   1   4917.1 7272.7

Step:  AIC=6735.7
doctorco ~ sex + age + income + levyplus + freepoor + freerepa +
    illness + actdays + hscore + cond

           Df Deviance    AIC
- freerepa  1   4381.0 6734.5
<none>          4380.1 6735.7
- cond      2   4384.2 6735.8
- age       1   4383.0 6736.5
```

```
- levyplus  1   4383.3 6736.9
+ agesq     1   4379.5 6737.1
- income    1   4385.0 6738.6
- freepoor  1   4386.8 6740.4
- sex       1   4388.0 6741.5
- hscore    1   4389.1 6742.7
- illness   1   4481.9 6835.4
- actdays   1   4917.1 7270.7

Step:  AIC=6734.53
doctorco ~ sex + age + income + levyplus + freepoor + illness + 
    actdays + hscore + cond

           Df Deviance    AIC
<none>          4381.0 6734.5
- levyplus  1   4383.4 6735.0
- cond      2   4385.5 6735.0
+ freerepa  1   4380.1 6735.7
+ agesq     1   4380.3 6735.8
- income    1   4386.7 6738.2
- age       1   4387.1 6738.7
- freepoor  1   4389.1 6740.6
- sex       1   4389.5 6741.0
- hscore    1   4390.2 6741.8
- illness   1   4482.7 6834.2
- actdays   1   4917.6 7269.2
```

```r
summary(step_model)
```

```
Call:
glm(formula = doctorco ~ sex + age + income + levyplus + freepoor + 
    illness + actdays + hscore + cond, family = poisson, data = dvisits)

Coefficients:
                         Estimate Std. Error z value Pr(>|z|)    
(Intercept)             -2.089063   0.100811 -20.723  < 2e-16 ***
sex                      0.162000   0.055824   2.902  0.00371 ** 
age                      0.355131   0.143196   2.480  0.01314 *  
income                  -0.199806   0.084328  -2.369  0.01782 *  
levyplus                 0.083689   0.053544   1.563  0.11805    
freepoor                -0.469596   0.176360  -2.663  0.00775 ** 
```

```
illness                    0.186101   0.018260  10.191  < 2e-16 ***
actdays                    0.126611   0.005029  25.177  < 2e-16 ***
hscore                     0.031116   0.010065   3.092  0.00199 **
condno_limited_activity  0.121100   0.066389   1.824  0.06814 .
condlimited_activity     0.158894   0.081762   1.943  0.05197 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 5634.8  on 5189  degrees of freedom
Residual deviance: 4381.0  on 5179  degrees of freedom
AIC: 6734.5

Number of Fisher Scoring iterations: 6
```

```r
# Predict for all people
dvisits$predicted_visits <- predict(step_model, type = "response")

# View top predicted individual
top_person <- dvisits[which.max(dvisits$predicted_visits), ]
top_person
```

```
     sex  age  agesq income levyplus freepoor freerepa illness actdays hscore
2404   1 0.72 0.5184   0.25        1        0        0       5      14      9
     chcond1 chcond2 doctorco nondocco hospadmi hospdays medicine prescrib
2404       0       1        0        0        0        0        4        4
     nonpresc age_group doctorco_plot             cond predicted_visits
2404        0       65+             0 limited_activity         4.501247
```

The strongest predictors of increased doctor visits were a higher number of recent illnesses, more days of activity limitation due to illness or injury, and a higher score in the health questionaire (high score indicates bad health) (hscore). Being female, older, and having a chronic condition (especially one that limits activity) were also associated with higher predicted visit counts. However, higher income was associated with fewer doctor visits, and being covered under government care due to low income (freepoor) was also negatively associated. Overall, the person predicted to visit the doctor the most would be someone who is female, older, in poor health (high illness count, high actdays, high hscore), and has a chronic condition that limits activity.

(f) (5pts)
For the last person in the dataset, compute the predicted probability distribution for their visits to the doctor, i.e., give the probability they visit 0, 1, 2, etc. times.

```
# Extract the last person
last_person <- dvisits[nrow(dvisits), ]

# Predicted mean (lambda) for Poisson distribution
lambda_hat <- predict(step_model, newdata = last_person, type = "response")
lambda_hat
```

```
     5190
0.1520842
```

```
# Values from 0 to 9 visits
k_vals <- 0:9
probs <- dpois(k_vals, lambda_hat)

# Create a data frame
prob_table <- data.frame(Visits = k_vals, Probability = round(probs, 4))
print(prob_table)
```

```
   Visits Probability
1       0      0.8589
2       1      0.1306
3       2      0.0099
4       3      0.0005
5       4      0.0000
6       5      0.0000
7       6      0.0000
8       7      0.0000
9       8      0.0000
10      9      0.0000
```

The predicted Poisson mean for the last person in the dataset is 0.152. Based on this, the most likely outcome is 0 visits, with approximately 86% probability. There's about a 13% chance of 1 visit, and the probability of more than 2 visits is nearly negligible. This reflects a low expected rate of healthcare utilization for this individual under the fitted model.

(g) (10pts)
Tabulate the frequencies of the number of doctor visits. Compute the expected frequencies of doctor visits under your most recent model. Compare the observed with the expected frequencies and comment on whether it is worth fitting a zero-inflated count model.

```r
# Observed Frequencies
table(dvisits$doctorco)
```

```
   0    1    2    3    4    5    6    7    8    9 
4141  782  174   30   24    9   12   12    5    1 
```

```r
# Expected Frequencies
lambda_all <- predict(step_model, type = "response")

expected_counts <- sapply(0:9, function(k) sum(dpois(k, lambda_all)))
names(expected_counts) <- 0:9
round(expected_counts, 1)
```

```
     0      1      2      3      4      5      6      7      8      9 
4013.6  928.3  168.0   45.5   18.9    8.8    4.0    1.7    0.7    0.3 
```

```r
# Number of zeros in observed vs. expected
obs_zeros <- sum(dvisits$doctorco == 0)
exp_zeros <- round(expected_counts["0"], 1)

cat("Observed zeros:", obs_zeros, "\n")
```

```
Observed zeros: 4141
```

```r
cat("Expected zeros:", exp_zeros, "\n")
```

```
Expected zeros: 4013.6
```

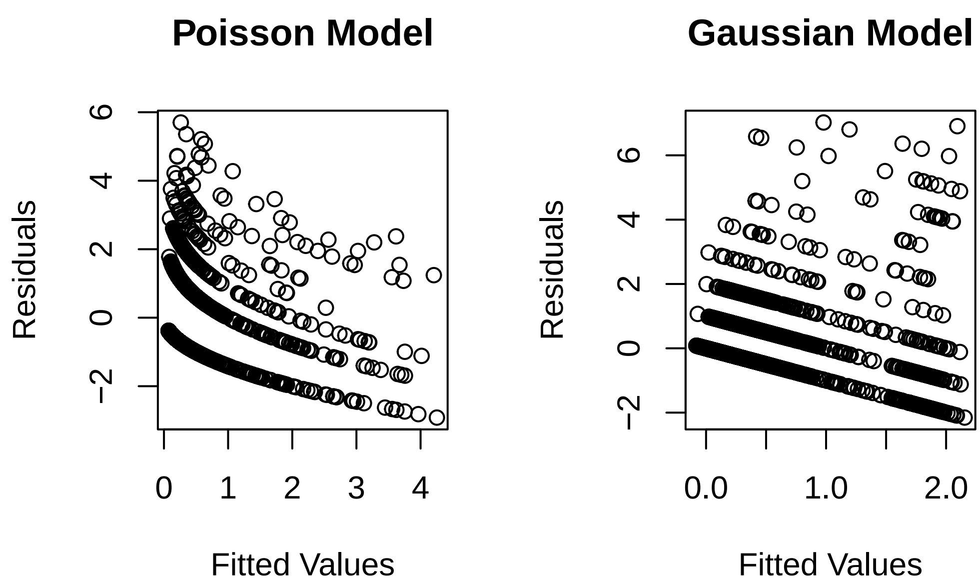
(h) (15pts)
Fit a comparable (Gaussian) linear model and graphically compare the fits. Describe how they differ.

```r
# Gaussian linear model
lm_model <- lm(doctorco ~ sex + age + income + levyplus + freepoor +
               illness + actdays + hscore + cond, data = dvisits)

summary(lm_model)
```

```

Call:
lm(formula = doctorco ~ sex + age + income + levyplus + freepoor +
    illness + actdays + hscore + cond, data = dvisits)

Residuals:
    Min      1Q  Median      3Q     Max
-2.1543 -0.2584 -0.1440 -0.0434  7.0211

Coefficients:
                         Estimate Std. Error t value Pr(>|t|)
(Intercept)              0.036781   0.035794   1.028 0.304201
sex                      0.035574   0.021505   1.654 0.098137 .
age                      0.180024   0.055912   3.220 0.001291 **
income                  -0.061208   0.030522  -2.005 0.044975 *
levyplus                 0.024041   0.021235   1.132 0.257626
freepoor                -0.111650   0.051532  -2.167 0.030308 *
illness                  0.060148   0.008332   7.219 6.02e-13 ***
actdays                  0.103140   0.003656  28.213  < 2e-16 ***
hscore                   0.017064   0.005180   3.294 0.000994 ***
condno_limited_activity  0.005006   0.023709   0.211 0.832776
condlimited_activity     0.045319   0.035352   1.282 0.199921
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7138 on 5179 degrees of freedom
Multiple R-squared:  0.2017,    Adjusted R-squared:  0.2002
F-statistic: 130.9 on 10 and 5179 DF,  p-value: < 2.2e-16
```

```r
par(mfrow = c(1, 2))

# Plot Poisson model fit
plot(full_model$fitted.values, resid(full_model), xlab = "Fitted Values",
     ylab = "Residuals", main = "Poisson Model")
```

```r
# Plot Gaussian model fit
plot(lm_model$fitted.values, resid(lm_model), xlab = "Fitted Values",
     ylab = "Residuals", main = "Gaussian Model")
```



Sex and age are no longer statistically significant in the Gaussian model, while they were in the Poisson model. While both models produced similar patterns in predicted values, the linear model assumes normally distributed errors and constant variance, which are inappropriate for count data like doctor visits.

0.4 Q4. Uniform association (20pts)

For the uniform association when all two-way interactions are included, i.e.,

$$\log \mathbb{E}Y_{ijk} = \log p_{ijk} = \log n + \log p_i + \log p_j + \log p_k + \log p_{ij} + \log p_{ik} + \log p_{jk}.$$

Proof the odds ratio (or log of odds ratio) across all stratum k

$$\log \frac{\mathbb{E}Y_{11k}\mathbb{E}Y_{22k}}{\mathbb{E}Y_{12k}\mathbb{E}Y_{21k}}$$

is a constant, i.e., the estimated effect of the interaction term "i:j" in the uniform association model